← MATH 401 LEC 11809 Spring 2021 Owens, Spring 2021

Section 5.5 Homework (Homework)

INSTRUCTOR
Rocio Owens
American River College, CA

Current Score

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
POINTS	1/1	4/4	1/1	1/1	1/1	1/1	1/1	1/1	5/5	1/1	1/1	1/1	1/1	1/1	1/1	1/1	1/1

TOTAL SCORE

24/24	100.0%

Due Date Past Due

THU, JAN 21, 2021
11:59 PM PST

Request Extension

Assignment Submission & Scoring

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

The due date for this assignment has passed.

Your work can be viewed below, but no changes can be made.

Important! Before you view the answer key, decide whether or not you plan to request an extension. Your Instructor may not grant you an extension if you have viewed the answer key. Automatic extensions are not granted if you have viewed the answer key.

Request Extension

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1. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.001.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the integral by making the given substitution. (Use C for the constant of integration.)

$$\int \cos(8x)\,dx, \quad u = 8x$$

$$sin(8x)8+C$$

Need Help? Read It

2. **[4/4 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.003.MI.SA.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

This question has several parts that must be completed sequentially. If you skip a part of the question, you will not receive any points for the skipped part, and you will not be able to come back to the skipped part.

Tutorial Exercise

Evaluate the integral by making the given substitution.

$$\int x^2\sqrt{x^3+8}\,dx, \quad u = x^3 + 8$$

Step 1

$du =$
$$3x2$$

We know that if $u = f(x)$, then $du = f'(x)\,dx$. Therefore, if $u = x^3 + 8$, then ✔ dx.

Step 2

If $u = x^3 + 8$ is substituted into $\int x^2\sqrt{x^3+8}\,dx$, then we have $\int x^2\,(u)^{1/2}\,dx = \int u^{1/2}\,x^2\,dx$.

We must also convert $x^2\,dx$ into an expression involving u.

$x^2\,dx =$
$$13

We know that $du = 3x^2\,dx$, and so ✔ du.

Step 3

Now, if $u = x^3 + 8$, then $\int x^2\sqrt{x^3+8}\,dx = \int u^{1/2}\left(\frac{1}{3}\,du\right) = \frac{1}{3}\int u^{1/2}\,du$.

This evaluates as

$\frac{1}{3}\int u^{1/2}\,du =$
$$29u(32)

✔ $+ C$.

Step 4

Since $u = x^3 + 8$, then converting back to an expression in x we get

$\frac{2}{9}\,u^{3/2} + C =$
$$29(x3+8)(32)+C

✔ .

You have now completed the Master It.

Need Help? Read It

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3. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.004.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the integral by making the given substitution. (Use C for the constant of integration.)

$$\int \sin^4(\theta)\cos(\theta)\,d\theta, \quad u = \sin(\theta)$$

$$sin5(θ)5+C

✔

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4. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.005.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the integral by making the given substitution. (Use C for the constant of integration. Remember to use absolute values where appropriate.)

$$\int \frac{x^3}{x^4 - 4}\,dx, \quad u = x^4 - 4$$

$$ln|x4−4|4+C

✔

Need Help? Read It

5. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.013.MI.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the indefinite integral. (Remember to use absolute values where appropriate. Use C for the constant of integration.)

$$\int \frac{dx}{7 - 6x}$$

$$-16\ln(|7-6x|)+C$$

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6. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.017.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the indefinite integral. (Use C for the constant of integration.)

$$\int \frac{e^u}{(4 - e^u)^2}\, du$$

$$14-eu+C$$

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7. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.021.MI.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the indefinite integral. (Use C for the constant of integration.)

$$\int \frac{(\ln(x))^{30}}{x}\,dx$$

$$ln(x)3131+C

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8. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.023.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the indefinite integral. (Use C for the constant of integration.)

$$\int \sec^2(\theta)\tan^5(\theta)\,d\theta$$

$$tan6(\theta)6+C

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9. **[5/5 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.034.MI.SA.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

This question has several parts that must be completed sequentially. If you skip a part of the question, you will not receive any points for the skipped part, and you will not be able to come back to the skipped part.

Tutorial Exercise

Evaluate the indefinite integral.

$$\int \frac{\cos(\pi/x^{28})}{x^{29}}\,dx$$

Step 1

We must decide what to choose for u.

If $u = f(x)$, then $du = f'(x)\,dx$, and so it is helpful to look for some expression for which the derivative is also present, though perhaps missing a constant factor.

We can re-write $\int \frac{\cos\left(\frac{\pi}{x^{28}}\right)}{x^{29}}\,dx = \int$ $\cos\left(\frac{\pi}{x^{28}}\right)$ $$1x29 dx.

We see that $\frac{\pi}{x^{28}}$ is part of this integral, and the derivative of $\frac{\pi}{x^{28}}$ is $$-28\pi x29 .

Step 2

If we let $u = \frac{\pi}{x^{28}}$, then $du = -\frac{28\pi}{x^{29}}\,dx$, and so we have

$$\int \cos\left(\frac{\pi}{x^{28}}\right)\frac{1}{x^{29}}\,dx = \int$$ $\cos(u)$ $$-128\pi du.

Step 3

$$-\frac{1}{28\pi}\int \cos(u)\,du =$$ $$-128\pi sin(u)$$

We have + C, and converting back to an expression in x gives us $$-128\pi sin(\pi x28)+C .

You have now completed the Master It.

Need Help? Read It

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10. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.039.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the indefinite integral. (Use C for the constant of integration.)

$$\int \frac{\sin(2x)}{42 + \cos^2(x)}\,dx$$

$$-ln(|cos(2x)+85|)+C$$

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11. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.044.MI.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the indefinite integral. (Use C for the constant of integration.)

$$\int \frac{x^7}{1 + x^{16}}\,dx$$

$$18arctan(x8)+C$$

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12. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.045.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the indefinite integral. (Use C for the constant of integration.)

$$\int \frac{9 + 9x}{1 + x^2}\,dx$$

$$9\arctan(x)+92\ln(|1+x2|)+C$$

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13. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.055.**

MY NOTES ASK YOUR TEACHER

Evaluate the definite integral.

$$\int_0^1 \sqrt[3]{1 + 7x}\,dx$$

$4528

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14. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.059.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the definite integral.

$$\int_1^2 \frac{e^{1/x^4}}{x^5}\,dx$$

$$e-e(116)4$$

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15. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.063.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the definite integral.

$$\int_0^{13} \frac{dx}{\sqrt[3]{(1+2x)^2}}$$

$$3

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16. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.067.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate the definite integral.

$$\int_8^9 x\sqrt{x-8}\,dx$$

$$8615

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17. **[1/1 Points]** DETAILS PREVIOUS ANSWERS **SCALCET8 5.5.077.**

MY NOTES ASK YOUR TEACHER PRACTICE ANOTHER

Evaluate $\int_{-5}^{5} (x+4)\sqrt{25-x^2}\,dx$ by writing it as a sum of two integrals and interpreting one of those integrals in terms of an area.

$$50\pi

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